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Cardiopulse

Sudden cardiac arrest at sporting events

Accurate first responder performance and feasible medical emergency plans are crucial to prevent fatal cardiac outcomes says Dr Christian Schmied

With the summer Olympics in Brazil recently finished and the football season soon to start in Europe, the topic of sudden cardiac death in sports, particularly at mass events comes into focus once more. It is all the more important after the tragic deaths of two professional football players in Romania and Brazil earlier this year. In one case, investigations have been started following rumors of massive omissions in the emergency medical care at the sport venue site and on the way to the hospital.

Background

Fatal cardiac events related to exercise and sports are a particular tragedy as it may strike presumably physically healthy young individuals. By applying current recommendations for cardiac screening of physically active individuals most of the incidences can be prevented. However, an accurate clinical assessment of the individuals is crucial and it has to be noted that the majority of athletes, particularly in amateur- and hobby-sports do not undergo regular cardiac assessments. Furthermore, it can be estimated that ~10% of sudden cardiac arrests (SCAs) cannot be prevented by pre-participation screening.

Considering all these facts, it is clear that measures of secondary prevention and 'rescue on the field' are of eminent importance. The use of automated external defibrillators (AEDs) significantly advances the treatment and prognosis of out-of-hospital SCA and has emerged and refined cardiopulmonary resuscitation (CPR) procedures.

In addition, there is another important fact that should be taken into account: It has been impressively demonstrated that spectators at sporting events incur a relevant increased risk for cardiac events during competitions of their athletes and teams. During the football world cup in Germany 2006, it could be demonstrated that viewing a match more than doubles the risk of an acute cardiovascular event.

Current situation and setup in sports arenas

Large sports stadiums often gather many thousands of spectators, which includes individuals with an increased risk for cardiovascular events. Thus, we have to be aware that fatal cardiovascular events may not occur just in athletes but also in spectators or other individuals involved in the game. In major European football stadiums, the incidence of SCA has been estimated to be 1–2 per 5–600 000 spectators.

The 'ARENA' study highlighted various items to improve in stadiums and demonstrated quite a high proportion of inadequacies regarding cardiovascular safety programmes in major European soccer arenas. As many as 28% of the participating clubs did not have an AED available in their stadiums, and 36% did not have a documented medical action plan (MAP) describing cardiovascular safety procedures. Furthermore, 41% of clubs reported a transportation time from the arena to the nearest hospital in excess of 5 min, and a substantial proportion of these clubs (25%) did not report the availability of an AED. These acts are alarming, particularly as the National Association of Emergency Medical Services Physicians recommends an accurate emergency concept including a MAP with and AED on the pitch as a blueprint for delivering emergency medical care at mass gathering events (e.g. >1000 persons). Moreover, in the USA, AEDs are recommended in large sports facilities and gyms, and also in the school setting, when the transport time to defibrillation exceeds the critical threshold of 5 min.

The Fédération Internationale de Football Association study and its consequences

In 2013, the Fédération Internationale de Football Association (FIFA) and its medical research and assessment center (F-MARC), evaluated standards of first responder measures on the pitch in its worldwide member associations with the aim to improve the global cardiopulmonary resuscitation concepts in football. The results were worrisome: Only approximately two-thirds of the associations provided regular training in BLS/ACLS and implemented standards as MAPs or had appropriate equipment (including an AED) during games of their national teams. In training sessions, the situation was even worse. And there was another important finding: on average, the education and practical BLS/ACLS skills of other involved persons (e.g. coaches, athletic trainers, groundkeepers, and referees) were clearly insufficient.

As a consequence, the FIFA created a detailed plan to improve these situations. The so-called 11 steps to prevent sudden cardiac death in football summarize screening, emergency planning, and management protocols for SCA. Together with the FIFA Medical Emergency Bag and AED the written, multi-language guide entitled 'Prevention and Management of Sudden Cardiac Arrest in Football', were provided to each of the 209 football associations. This concept was recommended as a global strategy in football.

The FIFA 11 Steps to prevent sudden cardiac death in football

Prevention
1. PCMA (pre-competition medical assessment)—player medical history (PMH), family history and physical examination
2. ECG—12-lead, resting, supine; initially + every year
3. Echocardiography—where necessary and at least once in the early career, exercise test where necessary (e.g. athletes >35 years old)

Planning + protocol
4. Training and equipment
A. CPR + AED training yearly for team staff and referees undertaken
B. FIFA medical emergency bag available and checked
C. Emergency medical plan: roles and responsibilities allotted; on field response practiced and rehearsed at least once annually
D. Field-of-play medical team qualifications + logistics confirmed
E. Ambulance location and logistics confirmed

Play the game + pre-game timeout
5. FIFA medical emergency bag with AED in position and checked
6. Field-of-play medical team in position (games)
7. Ambulance, fully functional, in position (games)

Performance of the emergency medical plan
8. Immediate recognition of collapsed player
A. Assume SCA if collapsed and unresponsive
B. Seizure activity and/or agonal respirations implicate SCA
9. Activation of emergency medical plan
10. Early CPR and AED application
A. Start chest compressions
B. Retrieve, apply and use AED as soon as possible
11. Early planned transition to advanced life support

Adapted from Dvorak J, Kramer E, Schmied C, Drezner, J, et al. Br J Sports Med 2013;47:1199–1202

Current recommendations and future perspectives

Prevention still remains better than any therapy and thus, a tailored individual cardiac screening of athletes involved in competitive sports should be considered, particularly in so-called amateur athletes. However, when the aim is to improve the performance of rescue issues directly on the field ('secondary prevention'), the first potential first responders have to be identified. Although teammates, athletic trainers, and coaches are mostly involved, there are other persons such as physicians, officials, and other staff or administrators that should be integrated in an MAP and should therefore be educated in the recognition of cardiac emergencies, CPR, AED use and have specific knowledge of the location of the nearest AED.

An approved communication plan should be installed at all venues and be integrated with the established local emergency medical system. The action plan further requires approved transportation routes for emergency vehicles to access and leave the specific location and hospitals that can be reached within adequate time.

The European Society of Cardiology provides recommendations for a consistent level of emergency cardiac care at sports venues and these recommendations are expected to be implemented by major sporting bodies.

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Pre-participation athlete screening: there is no 'one size fits all'

Dr Aaron Baggish discusses the US recommendations with selective use of electrocardiography in screening

Sudden cardiac death during sport is a rare but tragic phenomenon with wide reaching implications. The loss of life among young and ostensibly healthy people with occult cardiovascular disease devastates families, peer groups, and the community at large. Increasingly detailed media coverage of these tragedies often have the unanticipated effect of embedding fear into the general public which overshadows the numerous health benefits of participation in organized competitive athletics. Consequently, cardiovascular specialists worldwide, particularly those who focus on the care of athletes through the emerging field of sports cardiology, have long embraced a common mission to reduce the incidence of sudden cardiac death. The importance of maximizing the safety of athletic participation and therefore minimizing the risk of adverse cardiovascular events is not controversial. How to do so is less straightforward.

Pre-participation cardiovascular disease screening is one of several key components of a comprehensive strategy designed to protect athletes from sudden cardiac death. The primary objective of pre-participation screening is to identify cardiovascular conditions that increase the risk of sudden and unexpected collapse. Regardless of how screening is conducted, it is most often rationalized on the basis that it affords an opportunity to identify the small minority of athletes with high-risk cardiac conditions such that they can be withdrawn from sport participation or medically/surgically treated to reduce risk to an acceptable basis. In 1996, an expert panel convened by the